Guardians of the Grapevine`

Team: Guardians of the Grapevine

Clients: Cornell CALS Extension / E&J Gallo Winery / National Grape

- List of design questions: what do you want to test with the mock-up before moving to the actual prototype?

1) Do the dimensions and sizes make sense?
2) Where can we place the components, especially the battery?
3) What can we use to zap flies, i.e. what can produce UV light-based electric grids?

- Timeline: create a timeline starting from the final exhibition deadline, and your first prototype deadline. Identify where along the timeline you need to make key decisions (e.g. ordering material). Create this on a flipchart.

1) Week 1 2/24: Create timeline and questions
2) Week 2 3/3: Build mock-up and order components
    a) Begin cardboard prototype
    b) Check prototype for faults
    c) Start designing/cadding idea
3) Week 3 3/10: Build prototype and identify weaknesses
4) Week 4 3/17: Plan out model and order components
    a) Order final components
5) Week 5 3/24: Build actual model
6) Week 6 4/7: Testing and identify weaknesses
    a)
7) Week 7 4/14: Fix model using testing results
    a) Build final version of model
8) Week 8 4/21: More testing and identifying weaknesses
9) Week 9 4/28: Present!

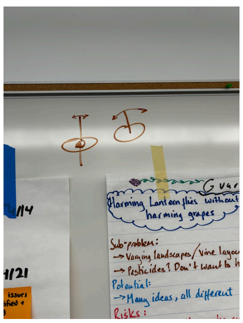
    Loose ends Wrapped up

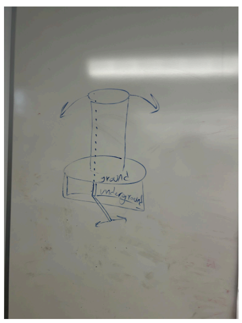
    Final Deadline - Everything must be done

Documentation (Henry)

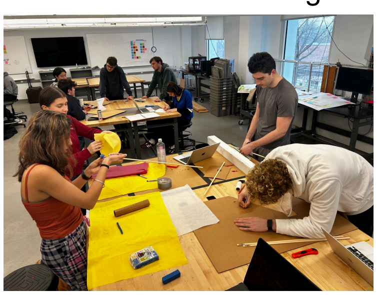
First we started putting up our posters on the whiteboard to visualise

Next, we had a little debate over the design and drew out our separate ideas to decide which path to take

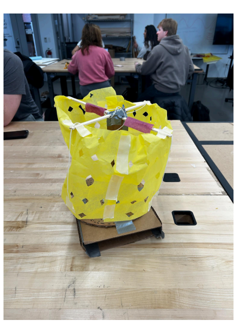
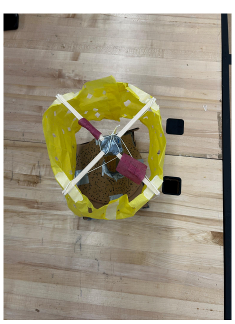


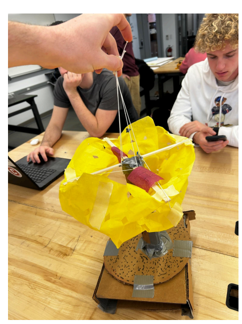


We decided on the design and got to building



This is the final design, note the strings that lift up the cage and the box underneath so we can make the stem vibrate.




Q&A (Katia)

1. **Do the dimensions and sizes make sense?**
   The proposed dimensions generally make sense as a starting point for the design because they reflect both the physical constraints of a vineyard and the functional needs of the device. A tower height of approximately 2.5 meters is reasonable because it allows the device to be tall enough to attract spotted lanternflies while also keeping active components elevated and away from the grape clusters. This height also provides enough vertical space to separate the different parts such as the vibration mechanism and the electrical grid. However, the height must still be tested because a tall structure could create stability issues if the base is not heavy or wide enough. The proposed tower width of 0.15 meters is beneficial because it keeps the device narrow enough to fit between vineyard rows without obstructing workers or equipment. At the same time, this width may be somewhat tight once all internal components such as wiring and structural supports are installed. The cage dimensions of approximately one meter in length and a radius of 0.25 meters appear reasonable because they create a capture area that is large enough to intercept flies without becoming too bulky. These dimensions also help ensure that the device does not interfere with the vines themselves. The material thickness estimate of about 2.5 millimeters also seems logical because it balances durability with weight, which is important for an outdoor structure that must remain portable, especially if different parts of the vineyard are more targeted by the pests. The total weight estimate of roughly twenty kilograms is appropriate because it is heavy enough to prevent the tower from easily tipping while still being light enough for a team to move. Overall, the sizes and dimensions are reasonable for an early design estimate, but they should still be validated through the mock-up and prototype stages.
2. **Where can we place the components, especially the battery?**
   The placement of components is very important because the internal layout of the device affects stability. The battery should most likely be placed near the bottom of the tower because batteries are typically one of the heaviest components in small mechanical systems. Placing the battery low helps lower the center of gravity, which improves the overall stability of the device and reduces the risk of the tower tipping over in wind or during operation. A lower battery position also makes maintenance easier because the battery can be accessed without needing to climb or disassemble upper sections of the structure. Ideally, the battery would be enclosed in a protected compartment at the base of the tower where it is shielded from rain, dirt, and accidental contact. This compartment should also allow for ventilation if necessary while still keeping the battery secure. The electric grid components should be located higher up in the active zone where the flies are meant to gather. This positioning allows the attraction mechanism to be visible and effective while keeping the electrical elements contained within the protective cage. The vibration component should be at the bottom where their motion can effectively travel through the structure without destabilizing the tower. Wiring should ideally run through the inside of the tower to protect it from weather and physical damage. Organizing the components vertically within the tower allows each part to operate without interfering with the others. Overall, placing the battery at the bottom and the attraction and zapping

mechanisms higher in the structure should provide a stable and secure way to place components.

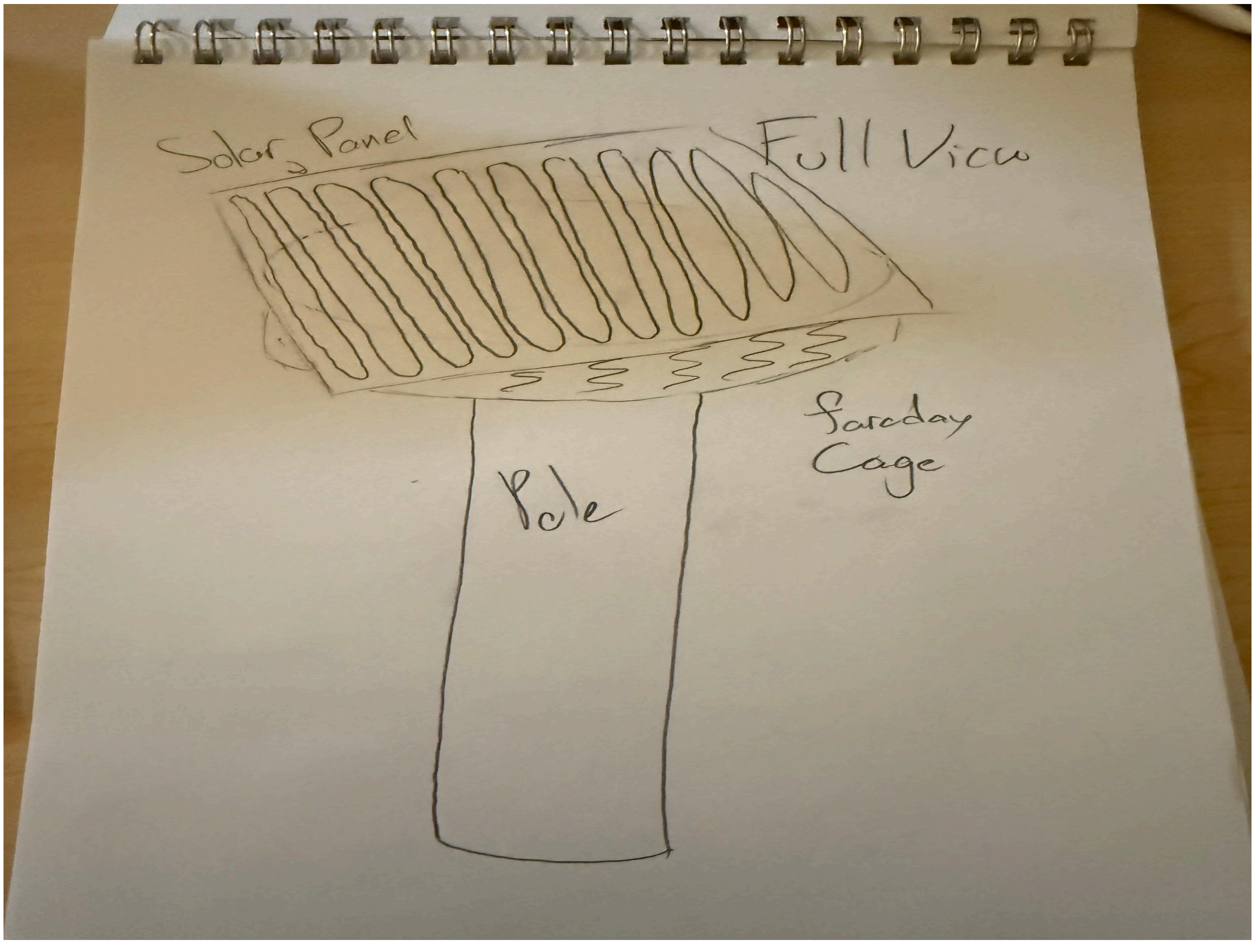
3. **What can we use to zap flies, i.e. what can produce UV light-based electric grids?** The most appropriate way to zap the flies in this design is to use an electrified cage made from conductive metal mesh arranged around the capture area, similar in concept to a Faraday-cage-style enclosure. Rather than relying on a single flat zapper panel, this design creates a surrounding active zone so that flies entering the trap are enclosed by the electrified structure. The cage could be made from metal screening, wire mesh, or parallel conductive wires shaped into a cylindrical form that matches the dimensions of the trap. This setup makes sense for the project because it combines containment and elimination in the same component, which is efficient for a compact vineyard device. A cage design also aligns well with the proposed radius and length estimates, since it can be built around the main trapping zone without making the device too wide. The conductive cage would need to be mounted securely and insulated from the rest of the tower so that electricity stays within the intended zapping region. It should also be surrounded by a nonconductive outer barrier or protective shell to prevent accidental contact by vines or grapes. Because the system will be outdoors, the cage must also be protected from the weather and debris that could interfere with performance or create safety issues. The main advantage of this approach is that it gives the flies multiple directions from which they can encounter the electrified surface instead of requiring them to hit one narrow panel. Overall, the best answer is that the flies can be zapped using a cylindrical conductive wire cage that acts as an electrified surrounding barrier inside the trap.

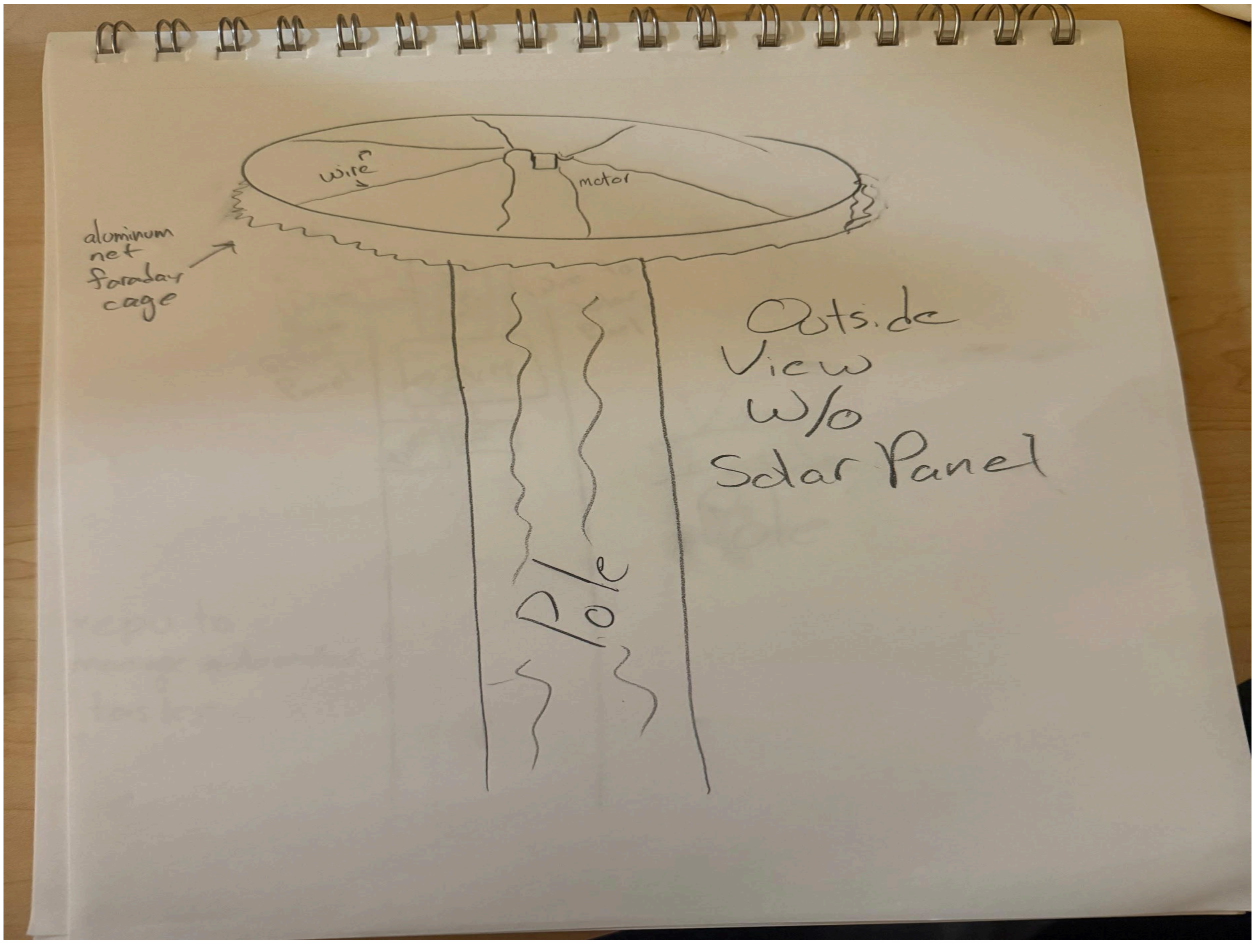
Design Estimates (Ash)

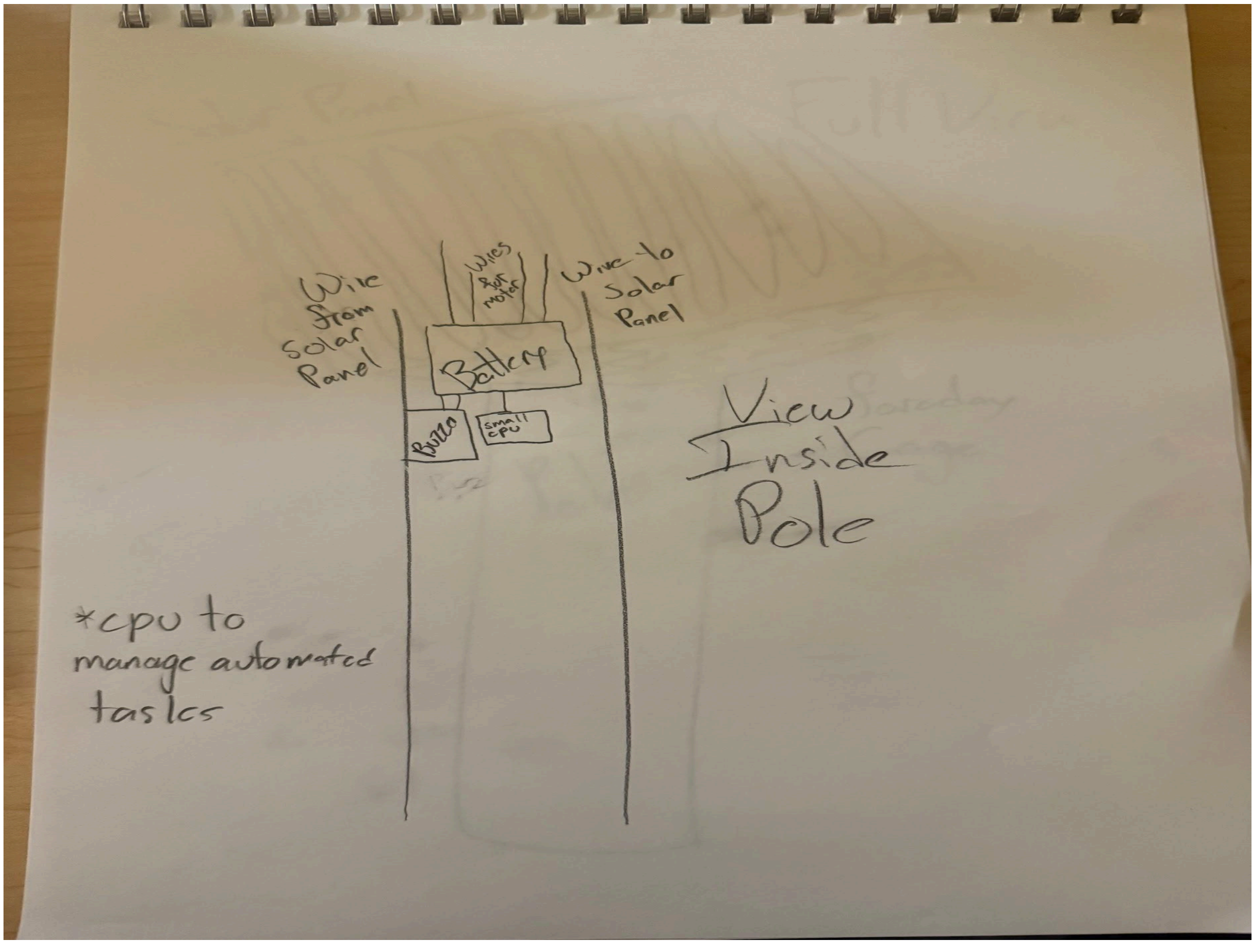
Dimension Name	Dimension Estimate	Derivation
Tower Length	2.5 m	Tall enough to attract the flies and a little extra height for safety and the cage so around 2.5-3 m would be feasible
Tower Width	0.15 m	Should be wide enough to hold the battery and the vibrating parts but be small enough to fit between the vineyard rows so 0.15 m seems like a good fit
Vibration Frequency	60 Hz	Lecture Speaker said that Spotted Lanternflies are attracted to 60 Hz frequency
Cage Length	1 m	Should be able to catch the flies when dropped without

		harming the grapes, so this seems like a reasonable estimate
Cage Radius	0.25 m	Should be large enough to have multiple flies inside but not so large that the voltage becomes unsafe so 0.25 m seems reasonable
Battery Voltage	12 V	Used online research and found that 12 V is typical of other small outdoor DC systems but it’s strong enough to power the zapper safely
Tower Material Thickness	2.5 mm	Light but strong enough to resist bending stress and most materials are 2-3 mm thick
UV Grid Power	5 W	Low power UV LEDs attract bats and flies in local testing areas according to online research so 5-10 W seems like a good idea
Total Weight	20 kg	Includes everything and should be portable so not too light but still strong enough not to blow away in the wind, so 20 kg seems reasonable

Fabrication and Assembly Analysis (James + Skyler)


Solar Panel
Full View
Faraday
Cage
Pole


wire
motor
aluminum net faraday cage
Outside View w/o Solar Panel
Pole



Wire from Solar Panel
Wires for motor
Wire to Solar Panel
Battery
Buzza
small cpu
View Inside Pole
*cpu to manage automated tasks